1 Derivatives

The derivative of a function f at a point a measures how quickly $f(x)$ changes when x is close to a. It is defined as the limit

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{1}$$

whenever that limit exists. A function with a derivative at every point of an interval is called differentiable there, and its graph has no corners on that interval.

1.1 Rules for derivatives

Three rules cover most of the functions met in this course:

- the power rule, $\frac{d}{dx} x^n = n x^{n-1}$ for every real n;
- the product rule, which also works
  - for three factors at once, and
  - for any number of factors by induction;
- the chain rule for compositions.

Written out in full, the product and chain rules read

$$\frac{d}{dx}\big(uv\big) = u'v + uv', \tag{2}$$

$$\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big)\, g'(x). \tag{3}$$

2 Integrals

2.1 The fundamental theorem

If F is an antiderivative of a continuous function f on $[a, b]$, so that $F' = f$, then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

A Riemann sum with n strips of width Δx approximates the same area:

$$\sum_{i=1}^{n} f(x_i)\, \Delta x \approx \int_a^b f(x)\, dx. \tag{4}$$

2.2 Exercises

1. Compute $\int_0^1 x^2\, dx$ with the fundamental theorem.
2. Show that
   (a) $\int_0^\pi \sin x\, dx = 2$, and
   (b) the area under $y = e^x$ between $x = 0$ and $x = 1$ is $e - 1$.
3. Estimate the integral in the first exercise with four strips.

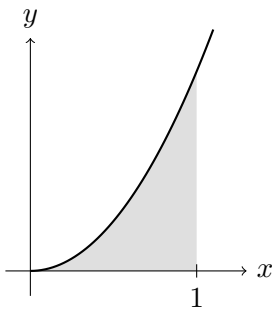

Figure 1: The area under $y = x^2$ between $x = 0$ and $x = 1$.

3 Continuity

The geometric series $\sum r^n$ converges exactly when $|r| < 1$, and the absolute value $|x|$ is continuous everywhere. The sign function

$$\operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

jumps at $x = 0$, so it has no derivative there.